

MSE603_Assignment 3

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Q1:

1. A bank has two sites at which checks are processed. Site 1 can process 11000 checks per day, and site 2 can process 7000 checks per day. The bank processes three types of checks: vendor, salary, and personal. The processing cost per check depends on the site as shown below. Each day, 6000 checks of each type must be processed.

Checks	Site (c)	
	1	2
Vendor	5	3
Salary	4	4
Personal	2	5

- a) (5points) Formulate a balanced transportation problem to minimize the daily cost of processing checks.
- b) (20points) Use the transportation simplex method to find the optimal solution of the problem in part a). (Use either the northwest corner or the minimum-cost methods to find an initial bfs)

Solution:

a) ①: Check this problem is balanced or unbalanced.

$$\text{Total Supply} = 11000 + 7000 = 18000$$

$$\text{Total demand} = 6000 \times 3 = 18000$$

Total Supply = Total demand, this is a balanced problem.

② Formulate as a balanced transportation problem.

Let: • x_{ij} = number of checks of type j processed at site i .

• $i = 1, 2$, Site 1, Site 2

• $j = 1, 2, 3$, Vendor, Salary, Personal.

$$\text{Min } Z = 5x_{11} + 3x_{21} + 4x_{12} + 4x_{22} + 2x_{13} + 5x_{23}$$

$$\text{s.t. } x_{11} + x_{12} + x_{13} = 11000$$

$$x_{21} + x_{22} + x_{23} = 7000$$

$$x_{11} + x_{21} = 6000, x_{12} + x_{22} = 6000, x_{13} + x_{23} = 6000.$$

$$x_{ij} \geq 0, \text{ integer } \forall i \in \{1, 2\}, \forall j \in \{1, 2, 3\}.$$

c_{ij}	Vendor	Salary	Personal	Supply
Site 1	$5(x_{11})$	$4(x_{12})$	$2(x_{13})$	11000
Site 2	$3(x_{21})$	$4(x_{22})$	$5(x_{23})$	7000
Demand	6000	6000	6000	18000

b) ① Find an initial BFS (Minimum-cost Method)

	Vendor	Salary	Personal	Supply
Site1	$\begin{matrix} 5 \\ X_{11} \end{matrix}$	$\begin{matrix} 4 \\ X_{12}=5000 \end{matrix}$	$\begin{matrix} 2 \\ X_{13}=6000 \end{matrix}$	$11000 - 6000 = 5000$ $-5000 = 0$
Site2	$\begin{matrix} 3 \\ X_{21}=6000 \end{matrix}$	$\begin{matrix} 4 \\ X_{22}=1000 \end{matrix}$	$\begin{matrix} 5 \\ X_{23} \end{matrix}$	$7000 - 1000 = 6000$ $-6000 = 0$
demand	6000 $-6000 = 0$	6000 $-5000 = 1000$	6000 $-6000 = 0$	

$$BV = \{X_{12}, X_{13}, X_{21}, X_{22}\} \quad Z = \cancel{55000} 54000$$

② Pricing out NBVs (determine the entering variable)

$$V_1 = 3 - 0 = 3 \quad V_2 = 4 - 0 = 4 \quad V_3 = 2 - 0 = 2$$

$U_1 = 0$	$\begin{matrix} 5 \\ X_{11} \end{matrix}$	$\begin{matrix} 4 \\ X_{12}=5000 \end{matrix}$	$\begin{matrix} 2 \\ X_{13}=6000 \end{matrix}$
$U_2 = 3 - 3 = 0$	$\begin{matrix} 3 \\ X_{21}=6000 \end{matrix}$	$\begin{matrix} 4 \\ X_{22}=1000 \end{matrix}$	$\begin{matrix} 5 \\ X_{23} \end{matrix}$

By setting $U_1 = 0$, we can know:

$$U_1 = 0, V_1 = 3, U_2 = 0, V_2 = 4, V_3 = 2$$

③ Calculate the reduced costs of BV.

$$\bar{C}_{ij} = U_i + V_j - C_{ij}$$

$$\bar{C}_{11} = 3 + 0 - 5 = -2 < 0$$

$$\bar{C}_{32} = 2 + 0 - 5 = -3 < 0$$

All the reduced costs are nonpositive,

so the current solution is optimal.

$$Z = 5X_{11} + 3X_{21} + 4X_{12} + 4X_{22} + 2X_{13} + 5X_{23}$$

$$= 5 \times 0 + 3 \times 6000 + 4 \times 5000 + 4 \times 1000 + 2 \times 6000 + 5 \times 0$$

$$= 54000$$

2. (25 points) A company is taking bids on four construction jobs. Three people have placed bids on the jobs. Their bids (in thousands of dollars) are given below (a * indicates that the person did not bid on the given job). Person 2 and Person 3 can do only one job, but person 1 can do as many as two jobs. Determine the minimum cost assignment of persons to jobs by using the Hungarian method.

Person	Job			
	1	2	3	4
1	50	46	42	40
2	51	48	44	*
3	*	47	45	45

Solution:

① Make it an standard assignment problem:

There are 4 jobs but only 3 people, and person 1 can do two jobs, so we duplicate person 1 and treat them as two identical workers.

Cost	job 1	job 2	job 3	job 4
person 1	50	46	42	40
person 1	50	46	42	40
person 2	51	48	44	∞
person 3	∞	47	45	45

② Row reduction:

Subtract the smallest element in each row from that row.

Subtract: row 1: 40, row 2: 40, row 3: 44, row 4: 45.

After row reduction:

$$C^{(1)} = \begin{bmatrix} 10 & 6 & 2 & 0 \\ 10 & 6 & 2 & 0 \\ 7 & 4 & 0 & \infty \\ \infty & 2 & 0 & 0 \end{bmatrix}$$

③ Column reduction:

Subtract the smallest element in each column from that column.

Subtract: col 1: 7, col 2: 2, col 3: 0, col 4: 0

After column reduction:

$$C^{(2)} = \begin{bmatrix} 3 & 4 & 2 & 0 \\ 3 & 4 & 2 & 0 \\ 0 & 2 & 0 & \infty \\ \infty & 0 & 0 & 0 \end{bmatrix}$$

We can't select 4 independent zeros.

Therefore, the algorithm must continue.

④ Cover all zeros:

We want to cover all zeros using the fewest row/columns.

$$C^{(2)} \begin{bmatrix} 3 & 4 & \textcircled{2} & 0 \\ 3 & 4 & \textcircled{2} & 0 \\ 0 & 2 & 0 & \infty \\ \infty & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{The smallest uncovered value is: } \theta = 2$$

⑤ Refine the matrix:

Subtract θ from every uncovered element.

Add θ to every element that lies at the intersection of 2 lines.

$$C^{(3)} \begin{bmatrix} 1 & 2 & \textcircled{0} & 0 \\ 1 & 2 & 0 & \textcircled{0} \\ \textcircled{0} & 2 & 0 & \infty \\ \infty & \textcircled{0} & 0 & 2 \end{bmatrix} \rightarrow \begin{array}{l} \text{Now we can find out } [1,3] \\ 4 \text{ independent zeros. } [2,4] \\ [3,1] \\ [4,2] \end{array}$$

⑥ Assign the jobs and calculate total cost.

Translate each chosen zero:

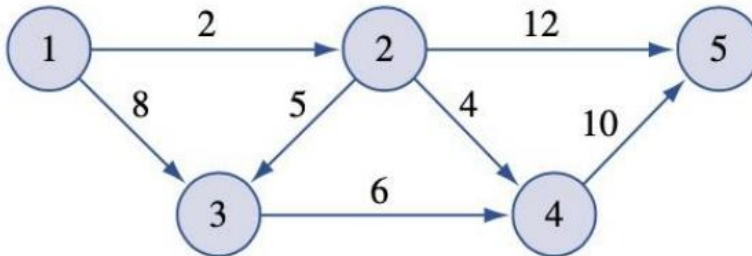
- $[1,3] \rightarrow \text{Person 1 does job 3.}$
- $[2,4] \rightarrow \text{Person 2 does job 4.}$
- $[3,1] \rightarrow \text{Person 3 does job 1.}$
- $[4,2] \rightarrow \text{Person 4 does job 2.}$

Total cost:

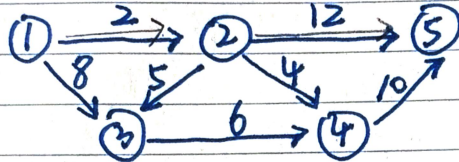
$$Z = 42 + 40 + 51 + 47 = 180$$

Q3:

3. (25 points) Find the shortest path from node 1 to node 5 using Dijkstra's Algorithm.



Solution:



Node: [1, 2, 3, 4, 5]

labels: [0*, ∞, ∞, ∞, ∞]

Temp label for node j (directly connected to node i)

= Min (node j's current label, node i's permanent label + c_{ij})

nodes: [1, 2, 3, 4, 5]

[0*, ∞, ∞, ∞, ∞]

[0*, 2*, 8, ∞, ∞]

[0*, 2*, 7, 6*, 14]

[0*, 2*, 7*, 6*, 14]

[0*, 2*, 7*, 6*, 14*]

2+5=7 < 8, ~~remain~~ change to 7.

6+10=16 > 14, remain 14.

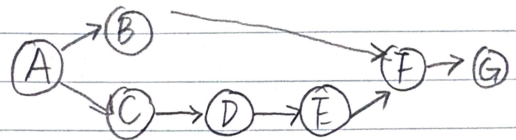
The shortest path is 1 → 2 → 5 with the total distance of 14.

4. (25 points) When an accounting firm audits a corporation, the first phase of the audit involves obtaining “knowledge of the business.” This phase of the audit requires the activities in the table. Draw the project network and determine the critical path for the network.

Activity	Description	Immediate Predecessors	Duration (Days)
A	Determining terms of engagement	—	3
B	Appraisal of auditability risk and materiality	A	6
C	Identification of types of transactions and possible errors	A	14
D	Systems description	C	8
E	Verification of systems description	D	4
F	Evaluation of internal controls	B, E	8
G	Design of audit approach	F	9

Solution:

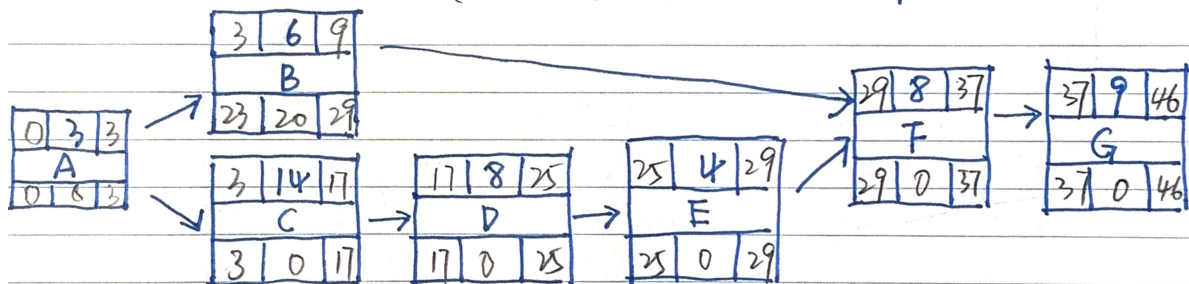
Activities	Days	Predecessors
A	3	—
B	6	A
C	14	A
D	8	C
E	4	D
F	8	B, E
G	9	F



Use this formulation:

ES	Duration	EF
Activity name		
LS	Slack	LF

Labels: Earliest start time (left), Earliest finish time (right), Latest start time (left), Latest finish time (right).



The critical path is: A → C → D → E → F → G